PHI 340: Philosophy of Science

Contact Info Travis McKenna
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434D Withers Hall

Time MW 1.30pm – 2.45pm
Location 318 Brooks Hall

Office Hours M: 12pm – 1pm
W: 9am – 10am

Description Modern science is one of the most successful intellectual projects of the last few hundred years. Physicist Richard Feynman once compared the accuracy of certain quantum theories to the ability to predict the distance between Los Angeles and New York within the width of a human hair. In part motivated by this astonishing success, philosophers began to dedicate a lot of energy to understanding modern science and its workings. Thus was born *the philosophy of science* as we know it today.

This course will serve as an introduction to a number of the key questions that philosophers have science have grappled with over the past hundred years (and beyond). What separates scientific knowledge from other ways that people claim to know about the world? How can we identify pseudoscience? How do scientific theories change? What does it mean for a scientific theory to be confirmed? Should we understand science as giving us a literally true description of reality? How do cultural values affect the way that science unfolds?

Goals By the end of this course, students will have developed the following:

- An in-depth understanding of *various core issues* in the philosophy of science, as well as an ability to explain some of the key positions with respect to these issues.
- The ability to relate some of the general claims made by philosophers of science to *more concrete and specific examples* drawn from various natural and social sciences.
- The ability to *present focussed objections to arguments*, which concentrate on either a particular premise or a particular inference in the argument's structure and suggest reasons to be dubious of it.
- The ability to *analyze the strength of different objections to an argument*, thinking about how an objection works, how central to the argument is the point that is challenged by the objection, and how easy it would be to modify the argument so as to avoid the objection.

Readings There is no textbook for this course. All the required readings will be made available via Moodle.

Assessment

Your final grade will be determined by the following assessments.

Quizzes	30%
Midterm Exam	25%
Final Exam	35%
Participation	10%

Quizzes: At the beginning of most of our Wednesday class sessions, the first 15 to 20 minutes will be devoted to a short quiz on the assigned reading for that day. These quizzes are designed to encourage consistent preparation, reward careful reading, and give you the opportunity to demonstrate your understanding before we discuss the material in detail. There will be a total of 12 quizzes over the course of the semester. Your best 10 results will count towards your grade.

The quizzes will be made up of multiple choice and short-answer questions focused on key arguments, concepts, and examples from the reading. All quizzes will be closed book and completed on paper. No electronic devices will be permitted during the quizzes. (Unless, of course, you have requested appropriate accommodations from NC State's Disability Resource Office.)

Midterm and Final: The midterm and final will take place on *September 24th* and *December 1st*, respectively. You will have an hour to complete these exams, which will consist of a combination of short-answer and short-essay questions, designed to assess your grasp of the core arguments and concepts. The midterm will focus on material from the first half of the course, where the final will cover the whole semester (though with an emphasis on the second half of the course).

Like the quizzes, the midterm and final will be closed book exams, completed on paper. No electronic devices will be permitted during the exams. (Once again, unless you have requested appropriate accommodations from NC State's Disability Resource Office.)

Participation: Developing the ability to discuss various positions and claims with your peers is an extremely important part of learning how to engage with ideas in a broadly philosophical fashion. A portion of your grade then will be determined by the way in which you go about developing and demonstrating these kinds of philosophical skills.

A wide variety of things will count as 'participation' for the purpose of your grade, including: asking questions in class, offering answers to questions posed either by me or your fellow students, coming to my office hours to discuss the course material, participating in regular in-class group activities, and so on.

Schedule

Module 1: Science and Pseudoscience
Sessions with quizzes are marked by an asterisk (*)

1/12 *Introduction and Housekeeping*
No required reading.

1/14	*What is Philosophy of Science?* Peter Godfrey-Smith, *Theory and Reality*, §1
1/19	*No Class (MLK Day)*
1/21	*The Demarcation Problem (1)** Karl Popper, "Science: Conjectures and Refutations"
1/26	*The Demarcation Problem (2)* Imre Lakatos, "Science and Pseudoscience"
1/28	*Cases of Pseudoscience ** Michael Ruse, "Creation-Science Is Not Science" Paul Thagard, "Why Astrology is a Pseudo-science"
2/2	*Is Demarcation Possible?* Larry Laudan, "The Demise of the Demarcation Problem"
	Module 2: Confirmation and Induction
2/4	*The Problem of Induction ** David Hume, selections from *A Treatise of Human Nature*
2/9	*Puzzles of Confirmation* Nelson Goodman, selections from *Fact, Fiction, and Forecast*
2/11	*Confirmational Holism ** Willard Van Orman Quine, "Two Dogmas of Empiricism"
2/16	*Bayesian Confirmation Theory* Wesley Salmon, "Rationality and Objectivity in Science or Tom Kuhn Meets Tom Bayes"
2/18	Midterm Exam
	Module 3: Theory Change and Observation
2/23	*Kuhn and Paradigms* Thomas Kuhn, selections from *The Structure of Scientific Revolutions* Optional Background: Peter Godfrey-Smith, *Theory and Reality*, §§5,6
2/25	*Observation (1) ** Dudley Shapere, "The Concept of Observation in Science and Philosophy"
3/2	*Observation (2)* Ian Hacking, "Do We See Through a Microscope?"

3/4	*Observation (3)* * Jerry Fodor, "Observation Reconsidered"
3/9	*Underdetermination (1)* Margaret Greta Turnbull, "Underdetermination in Science: What It Is and Why We Should Care"
3/11	*Underdetermination (2)* * Kyle Stanford, "Refusing the Devil's Bargain: What Kind of Underdetermination Should We Take Seriously?"
3/16	*No Class (Spring Break)*
3/18	*No Class (Spring Break)*
	Module 4: Scientific Realism
3/23	*Scientific Realism Overview* Peter Godfrey-Smith, *Theory and Reality*, §12
3/25	*Realism and the 'No Miracles' Argument* * Timothy Lyons, "Explaining the Success of a Scientific Theory"
3/30	*Antirealism* Bas van Fraassen, "Arguments Concerning Scientific Realism," from *The Scientific Image*
4/1	*The Pessimistic Metainduction* * Larry Laudan, "A Refutation of Convergent Realism"
4/6	*Selective Realism* Stathis Psillos, "Scientific Realism and the 'Pessimistic Induction"
4/8	*Structural Realism* * John Worrall, "Structural Realism: The Best of Both Worlds?"
4/13	*Against Selective Realism* Kyle Stanford, "Pyrrhic Victories for Scientific Realism"
4/15	*Entity Realism* * Ian Hacking, "Experimentation and Scientific Realism"
4/20	*Inference to the Best Explanation* Nancy Cartwright, "When Explanation Leads to Inference"
4/22	*Are Scientific Theories Even True?* Nancy Cartwright, "Do the Laws of Physics State the Facts?"

4/27	*Final Exam Review*

Academic Integrity

All of the assessments for this course will take place during class time. Nonetheless, you must ensure that all the work that you submit is your own. If you incorporate the ideas of others, cite those sources. Do not copy language too closely. Even when summarising and paraphrasing cited sources, you must use your own language and present the ideas in an original way. And obviously, do not collude or communicate with other students or access any electronic devices or notes during the in-class assessments. If you are found doing so, you will receive a mark of zero for that assessment.

Violations of academic integrity will be handled in accordance with the Student Discipline Procedures (NCSU REG 11.35.02).

Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, 2751 Cates Avenue, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (NCSU REG 02.20.01).